



Huy Nguyễn Thanh <bihuyz@gmail.com>

PE study to complete

12 thư

Ulas Bagci <ulas.bagci@northwestern.edu>

lúc 00:17 26 tháng 3, 2026

Đến: bihuyz <bihuyz@gmail.com>, Vanshali Sharma <vanshali.sharma@northwestern.edu>

Cc: vedatacicek <vedatacicek@gmail.com>, Gorkem Durak <gorkem.durak@northwestern.edu>, Halil Ertugrul Aktas <halilertugrul.aktas@northwestern.edu>, Monica Dou <monica.dou@northwestern.edu>

Dear Vanshali and Huy

TIMELINE: We should publish this in Media or TMI. Submission Deadline: 1 June 2026.**(Little bit long email, but read it, multiple times, and print it, have it a guideline for the work).**

Here are the roadmaps for the rest of the PE study.

1. **Vanshali**, can you share all the data and the code with Huy and his team (HUY, please include anyone from your team whom will help you).
 - a. **Vanshali**: please indicate how many data from Stanford, and how many from Turkey are in your hand to deliver to Huy?
 - b. **Vanshali**: Can you please tell simple properties of Stanford data? CT+EHR? Or CT+report? Do they have prognosis labels too or not? You can reply to this email and then include your responses under the questions with red fonts.
 - c. What was your initial results, **Vanshali** ? AUC, accuracy etc. can be good to report here.
 - d. What is the state-of-the-art method in diagnosis? What is AUC there and how many data they used?
 - e. What is the state-of-the-art method in prognosis? What is AUC there and how many data they used?
2. Plan for completion of the study (**HUY and team**, please radiological part always connect with Gorkem and Erto, and Cardiac related questions and PESI related clinical scores, Vedat will be there always):
 - a. You should treat this problem as a **staged, multimodal, anatomy-aware learning problem**, not as one single end-to-end model from day one.
 - b. There are now several relevant 3D CT foundation-model directions: CT-CLIP/CT-RATE paired 25,692 non-contrast chest CT scans with reports; CT-FM was pretrained on 148,000 CT scans with label-agnostic contrastive learning; and organ-separated CT foundation models are starting to appear. For our problem, those are useful starting points, but because our data are contrast-enhanced CTPA (or noncontrast? Gorkem, Vanshali please clarify) and you already have ~20K in-house image-report pairs

(Stanford data), I would use public CT FM weights only as initialization and then continue pretraining on our own CTPA data. That domain match is likely more important than using a generic public chest-CT model unchanged.

- i. <https://arxiv.org/pdf/2501.09001> CT-FM
- ii. <https://arxiv.org/pdf/2403.17834> CT-RATE
- iii. <https://arxiv.org/pdf/2601.00260> TotalFM (organ separated CT foundational model)
- iv. <https://aimi.stanford.edu/datasets/inspect-Multimodal-Dataset-for-Pulmonary-Embolism-Diagnosis-and-Prognosis> STANFORD INSPECT data. Vanshali has this data, Monica can join to analyze EHR+CT part of this data as a part of her phd.
- v. <https://aimi.stanford.edu/datasets/radfusion-multimodal-pulmonary-embolism-dataset> (RadFusion), not sure if this data is available.
- vi. Check all the available PE data in the literature (HUY+his team).
- c. For prognosis, I strongly recommend making the primary endpoint: 30-day mortality among confirmed acute PE patients. Then, if you want, add a secondary endpoint of all-comer 30-day mortality among everyone scanned. The reason is that whole-chest CT contains strong non-PE mortality signals such as pleural effusion, pericardial effusion, malignancy-related findings, interstitial lung disease, and ascites (GORKEM,ERTO please check); a classic CTPA risk score outperformed PESI partly because those broader thoracic findings carried prognostic information. **If you mix PE-negative and PE-positive patients into one main mortality task, the model may mostly learn “sick patient” rather than “PE severity.”**
 - i. <https://pubmed.ncbi.nlm.nih.gov/27591768/> 2016 study, with CT based mortality prediction. Old but good study.
 - ii. <https://pubmed.ncbi.nlm.nih.gov/16020800/> prognosis study, clinical.
- d. Use the 20K reports to create **silver labels** and to pretrain the image encoder. A prior NLP study on CTPA reports showed very strong performance for extracting PE presence, acute vs chronic, central PE, and subsegmental PE from reports. That makes report mining a very reasonable weak-supervision strategy for Stanford cohort.
 - i. <https://pubmed.ncbi.nlm.nih.gov/25117751/> related study on location, type etc PE.
 - ii. I would extract at least these report-derived labels:
 - iii. PE present / absent
 1. acute / chronic / uncertain
 2. central / lobar / segmental / subsegmental / saddle
 3. right-heart strain terms: RV enlargement, RV/LV ratio mention, septal bowing, reflux
 4. pleural effusion / pericardial effusion
 5. malignancy-related findings
 6. chronic lung disease / fibrosis / emphysema if present

- This label set is valuable because published PE work already shows benefit from multitask learning around PE properties and RV/LV information, and CT-based prognostic models use broader thoracic findings beyond clot presence alone.
- e. Build a shared 3D CTPA encoder with two pretraining signals on the 20K cohort: (1) self-supervised volume learning on all scans, (2) image-report alignment using the reports. A practical recipe is: initialize from a public 3D CT FM if available, then continue pretraining on your own CTPA, then fine-tune for diagnosis and prognosis.
 - i. This is the safest route because public chest-CT report-paired models are currently based largely on non-contrast CT, whereas your downstream signal lives in contrast-enhanced pulmonary angiography. That mismatch makes in-domain continued pretraining especially attractive.
 - f. **DIAGNOSIS MODEL.** For PE diagnosis, use a multitask model rather than only PE yes/no. Published PE multitask work has already shown that jointly predicting PE presence plus attributes like location and RV/LV-related information is feasible and competitive. Also include a strong non-foundation baseline such as a PENet-style end-to-end 3D model, because PENet showed solid internal and external AUROC on PE detection and will help you prove the added value of the FM route.
 - i. PENet: <https://pubmed.ncbi.nlm.nih.gov/32352039/>
 - ii. Multitask learning PE: <https://pubmed.ncbi.nlm.nih.gov/35906477/>
 - g. **PROGNOSIS MODEL.** For the 300–400 cases (**Vanshali will confirm how many we have**) with mortality and clinical variables, do parameter-efficient transfer from the pretrained encoder. In other words, do not fully train a huge 3D model from scratch on that small prognosis cohort. Freeze most of the encoder or use adapters/LoRA-like fine-tuning, then add:
 - i. an image branch from the pretrained CTPA encoder
 - ii. a clinical branch from a small MLP/TabNet-style tabular model
 - iii. a fusion block
 - iv. a prognosis head
 - v. Published PE studies increasingly show that multimodal fusion beats either modality alone. One 2023 PE study reported up to a 14% performance boost from combining volumetric CTPA and clinical data, and 2025 PE survival studies found multimodal CTPA+clinical (+PESI) models outperformed PESI alone.
 1. PESI models can be Learned from VEDAT.
 - vi. For clinical variables, use:
 1. raw values
 2. missingness indicators
 3. a “common-variable subset” analysis
 - vii. That way you can report:
 1. image only

2. clinical only
3. image + clinical
4. image + clinical + PESI/sPEI

This comparison matters because PESI is still the established clinical benchmark family for PE prognosis.

h. **ANATOMY COUNTERFACTUAL EXPERIMENTAL SUITE.** I have this idea before but none of you tested before. It is about the role and location of PE on heart etc. Here how it works. Use the same scan to generate multiple inputs:

- i. Full CTPA
- ii. Heart + mediastinum (ROI1)
- iii. Strict heart only (ROI2)
- iv. Exclude pulmonary trunk and main pulmonary arteries as much as possible. (ROI3)
- v. Pulmonary arteries only (ROI4)
- vi. Lungs with intrapulmonary vessels (ROI5)
- vii. Lung parenchyma only (ROI6)
- viii. Remove heart and, as much as feasible, central hilar vessels. (ROI7)
- ix. Random same-size crop / body-wall control (ROI8)

This is your leakage sanity check. Please get Gorkem/Erto help on this and segment as much as possible to segment everything for this part. That last control is important. If a random crop predicts too well, you likely have protocol or acquisition leakage rather than biology.

1. **What you can conclude from heart ROI, and what you cannot**

- a. A heart ROI predicting prognosis is biologically very plausible. CT-detected RV dilation is associated with increased 30-day mortality in acute PE, and automated RV/LV ratio measurement on CTPA is feasible and prognostic. So a heart-focused prognosis model would fit known pathophysiology. <https://pubmed.ncbi.nlm.nih.gov/24603813/>
<https://pubmed.ncbi.nlm.nih.gov/33459854/>
- b. A lung ROI predicting prognosis is also plausible. Beyond visible emboli, lung-side vascular changes and broader thoracic findings can matter: reduced small venous volume on CT has been associated with higher 30-day and 90-day mortality, and CT-based PE risk scores use non-cardiac chest findings such as pleural effusion, malignancy-related lesions, and chronic interstitial lung disease. <https://pubmed.ncbi.nlm.nih.gov/34544259/>
<https://pubmed.ncbi.nlm.nih.gov/27591768/>
- c. But for **diagnosis**, if “heart-only” works, you should be careful with the biological claim. That does **not** automatically prove cardiac anatomic remodeling from PE. It could reflect:

- i. secondary hemodynamic consequences,
- i. chamber opacification/contrast timing,
- i. reflux,
- i. visible central clot near the heart,
- i. protocol confounding

(I have talked this part with Gorkem, he was right about it, since we have no disease information like that in our cohort, we should be more careful but can still talk about this, considering that these are mentioned in the discussion).

To sum up, the paper can be called:

An anatomy-partitioned multimodal foundation model with counterfactual erasure.

Concretely:

- one shared CTPA encoder pretrained on image-report pairs
- organ-specific adapters for heart, pulmonary arteries, lungs
- report-derived auxiliary tasks per organ
- clinical fusion only at the prognosis stage
- teacher-student distillation from full-CT teacher to ROI-specific students
- paired counterfactual masking on the same scan

After reading all these papers and discussing with Vedat and Gorkem, then I concluded that this new method I am telling you guys is

- scientifically cleaner,
- it directly answers **where the predictive information lives**,
- and it aligns with emerging organ-separated and region-grounded CT foundation-model work (this work is from Hugo's group at Harvard, 148,000 CT pairs). Region-grounded chest CT datasets now exist with organ-level masks and sentence-region grounding, and an organ-separated CT FM preprint reported promising organ-wise generalization.

There is little similar framework I have noticed in the literature, not from PE field but from Machine Learning. It is called concept-learning or concept-bottleneck. I assume that Kayhan may've a paper on this problem. Anyways, let me tell you how it works, keep in mind this:

a) image → latent concepts

b) concepts = RV/LV ratio, septal bowing, centrality, embolic burden proxy, pleural effusion, pericardial effusion, vascular pruning proxy

c) concepts + clinical variables → prognosis

That gives you interpretability beyond saliency maps.

Summary of diagnosis experiments:

- report-only NLP baseline
- image-only full-CT baseline
- image-only full-CT with FM initialization
- multitask image model with report-derived auxiliary labels
- strict-heart model
- pulmonary-artery-only model
- lung-parenchyma-only model
- random-crop negative control

(STARD+AI principles will be used for evaluation, reporting for papers..natMedicine paper) <https://pubmed.ncbi.nlm.nih.gov/40954311/>

Summary of prognosis experiments:

- PESI / sPESI baseline
 - clinical-only baseline
 - image-only full-CT baseline
 - image + clinical fusion
 - image + clinical + PESI
 - strict-heart prognosis model
 - lung-only prognosis model
 - artery-only prognosis model
 - PE-positive-only primary analysis
 - all-comer secondary analysis
- (TRIPOD+AI principles will be used for evaluation, reporting for papers)
<https://pubmed.ncbi.nlm.nih.gov/38626948/>

After checking TRIPOD+AI and STARD+AI you will see that we have to follow some rules for evaluation and reporting the results, basically:

- split by **patient**, never by scan
- prefer a **temporal split** if available
- use repeated or nested CV for prognosis if the event count is small
- report AUROC **and** AUPRC
- report calibration for prognosis
- use bootstrap confidence intervals
- test ROI-model differences on the same held-out patients

Note: if data is not available fully, you can ignore some of the experiments.

PS. I included **Monica** as this is related to her too, some parts she can get benefit from Foundational Models and etc. Monica feel free to join the study from different aspects. Especially FM.

Let me know if you have any questions.

Thanks for participating the study. It will be a good one, solving an important problem that has never been solved in the same way before!

Ulas Bagci, Ph.D.,

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BOOK A MEETING WITH (ZOOM)

<https://calendly.com/ubo7517-ads/30min>

Ulas Bagci <ulas.bagci@northwestern.edu>

lúc 00:18 26 tháng 3, 2026

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I am sending again with Correct email of Vanshali.

[Ân văn bản trích dẫn]

Vanshali Sharma <sharmavanshali@gmail.com>

lúc 13:03 26 tháng 3, 2026

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Hi Dr. Ulas,

Sure, I will share the details. Please give me some time to gather everything and will get back to you.

Regards,
Vanshali

[Ân văn bản trích dẫn]

Ulas Bagci <ulas.bagci@northwestern.edu>

lúc 07:44 15 tháng 4, 2026

Đến: Vanshali Sharma <sharmavanshali@gmail.com>

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I need some updates from you folks.

Let me start initial steps to motivate you:0

Vanshali: please indicate the source of data (Stanford etc), and Turkey (how many), etc. And please deliver them to Huy and team.

Vanshali: data description you can little tell. Initial results please too.

Vanshali+Huy: SOTA Methods for diagnosis, and AUC typically?, SOTA methods for prognosis, AUC?

Huy: Staged, multimodal anatomy aware problem you should treat. Did you check CTFM-CTRATE-TOTAL-FM foundational model already?

Huy: Check all the available PE data in the literature apart from what we have with Vanshali.

Huy: Include Vi and others if you deem so.

For now. The rest is written below. I made it a document you can make it a document to go over one by one,

I send notion document so that you can follow the instruction and timeline there.

Ulas Bagci, Ph.D.

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Associate Professor (with Tenure), Feinberg School of Medicine,

Department of Radiology, Northwestern University, Chicago, IL, USA, 60611

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[Ân văn bản trích dẫn]

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Vanshali Sharma <sharmavanshali@gmail.com>

lúc 14:41 22 tháng 4, 2026

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Hi Dr. Ulas,

Please find my point-by-point responses to the questions in the email below.

Regards,
Vanshali

On Wed, Apr 15, 2026 at 5:14 PM Ulas Bagci <ulas.bagci@northwestern.edu> wrote:

I need some updates from you folks.

Let me start initial steps to motivate you:0

Vanshali: please indicate the source of data (Stanford etc), and Turkey (how many), etc. And please deliver them to Huy and team.

The Stanford **INSPECT dataset** consists of approximately 23K CT scans with both diagnosis and prognosis labels. Given its large size (approx. 1.8 TB), it would be preferable to download it directly from the original source (link provided in the next answer).

The following columns are present in their labels file.

impression_id, pe_positive_nlp, 1_month_mortality, 6_month_mortality, 12_month_mortality, 1_month_readmission, 6_month_readmission, 12_month_readmission, 12_month_PH, pe_acute, pe_subsegmentalonly, pe_positive, tte_mortality, is_censored_mortality, tte_readmission, is_censored_readmission, tte_PH, is_censored_PH, tte_Atelectasis, is_censored_Atelectasis, tte_Cardiomegaly, is_censored_Cardiomegaly, tte_Consolidation, is_censored_Consolidation, tte_Edema, is_censored_Edema, tte_Pleural_Effusion, is_censored_Pleural_Effusion

According to my records, the **Turkey dataset** contains approximately 700 scans and is currently available on the server. I will confirm the exact count with Monica. If Huy has access to the server, I can share the path directly. Otherwise, Monica can assist with downloading and sharing the data with him.

Additionally, I noticed that in the literature, the RSPECT dataset is commonly used for pre-training and is publicly accessible here: <https://www.kaggle.com/competitions/rsna-str-pulmonary-embolism-detection/data>.

Vanshali: data description you can little tell. Initial results please too.

INSPECT dataset has **CT scans + reports + EHR**, out of which CT scans and reports can easily be downloaded from <https://stanfordaimi.azurewebsites.net/datasets/151848b9-8b31-4129-bc25-cefdf18f95d8>.

EHR needs some special access using <https://stanford.redivis.com/datasets/dzc6-9jyt6gapt> and are standardized in the OMOP CDM schema. Earlier, I had access to the dataset, but the mapping file release was still pending at that time. Now that the mapping has been provided, my access via the Northwestern login is being denied.

I do have a few files downloaded from before, but I am not certain whether they constitute the complete EHR dataset. @Monica Dou do you have the access to this EHR data?

Results: Some of the selected results can be found in [this](#) sheet. Other results are present in [this](#) sheet (particularly first two sheet tabs). Please note that I obtained better results using the Turkey dataset.

Vanshali+Huy: SOTA Methods for diagnosis, and AUC typically?, SOTA methods for prognosis, AUC?

The INSPECT dataset paper reported 0.721 and 0.794 AUC for diagnosis and prognosis, respectively (using only scans). Other than that, there are a few papers on report generation that employ a classifier module and report its results separately (using the INSPECT dataset). Links to these papers are given below.

<https://www.sciencedirect.com/science/article/pii/S1361841525003329>

<https://www.nature.com/articles/s41746-025-01807-8.pdf>

The following kaggle solutions could be helpful:

Kaggle solutions for RSPECT: <https://www.kaggle.com/competitions/rsna-str-pulmonary-embolism-detection/leaderboard?tab=public>

For HU modifications (it can help improve results): <https://www.kaggle.com/competitions/rsna-str-pulmonary-embolism-detection/writeups/give-us-gold-12th-place-solution>

Some other details: https://docs.google.com/document/d/1s8jJUuWavIY1_BhG-1keEVSwnDswlWH3Dnb15VZm4Pw/edit?usp=sharing

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lúc 14:50 22 tháng 4, 2026

Thanh-Huy Nguyen, MSc
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Mellon
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[Ấn văn bản trích dẫn]

 **img-90e4e988-caa7-461e-a56f-b6df05338971**
14K

Mailsuite Reminder <reminders@mailsuite.com>
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Đến: bihuyz@gmail.com

lúc 14:50 23 tháng 4, 2026

⚠ Your email to vulanvi22@gmail.com has not been opened yet. Snooze for 24H, 48H or 72H (disable)

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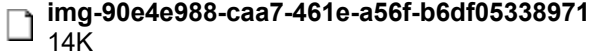
lúc 15:45 18 tháng 5, 2026

Huy, Vanshali, let us meet Wednesday morning please.

Huy: how is everything so far?
Ulas.

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[[Ấn văn bản trích dẫn](#)]



Vanshali Sharma <sharmavanshali@gmail.com>

lúc 05:20 19 tháng 5, 2026

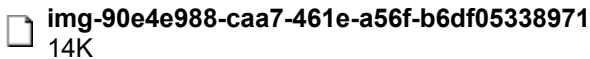
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Vanshali reacted via [Gmail](#)

[[Ấn văn bản trích dẫn](#)]



Huy Nguyễn Thanh <bihuyz@gmail.com>

lúc 09:25 20 tháng 5, 2026

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Hi Dat, Vi, Thien

This is our PE project description today.

Best,

--

Thanh-Huy Nguyen, MSc
Research Associate

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**Carnegie
Mellon
University**

Vào Thứ 5, 26 thg 3, 2026 vào lúc 00:17 Ulas Bagci <ulas.bagci@northwestern.edu> đã viết:

[[Ấn văn bản trích dẫn](#)]

Mail Delivery Subsystem <mailer-daemon@googlemail.com>
Đến: bihuyz@gmail.com

lúc 09:26 20 tháng 5, 2026



Không tìm thấy địa chỉ

Thư của bạn không được gửi đến **vanshali.sharma@northwestern.edu** vì không thể tìm thấy địa chỉ hoặc địa chỉ không thể nhận thư.

Phản hồi từ máy chủ từ xa là:

550 5.1.1 User Unknown

Final-Recipient: rfc822; vanshali.sharma@northwestern.edu
Action: failed
Status: 5.1.1
Remote-MTA: dns; chcsprrf12.ads.northwestern.edu. (165.124.29.102, the server for the domain northwestern.edu.)
Diagnostic-Code: smtp; 550 5.1.1 User Unknown
Last-Attempt-Date: Wed, 20 May 2026 06:26:01 -0700 (PDT)

 **noname**
5K

Ulas Bagci <ulas.bagci@northwestern.edu> lúc 12:38 25 tháng 5, 2026
Đến: bihuyz <bihuyz@gmail.com>, thienhaiquan123 <thienhaiquan123@gmail.com>, "vulanvi22@gmail.com" <vulanvi22@gmail.com>, Tien Dat Chung <chungtiendat8102000@gmail.com>
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The meeting link was here (video)

 [video1030737149.mp4](#)

For those the link does not work, I will try to send via WhatsApp

Ulas Bagci, Ph.D.

Senior Member, IEEE,
Associate Professor (with Tenure), Feinberg School of Medicine,
Department of Radiology, Northwestern University, Chicago, IL, USA, 60611
Director of Machine & Hybrid Intelligence Lab.
Courtesy Professor, Biomedical Engineering, and Electrical and Computer Engineering
Northwestern University, Evanston, IL, USA
Courtesy Professor, Center for Research in Computer Vision
University of Central Florida, Orlando, FL, USA

Steering Committee Member, Artificial Intelligence Resource at NCI / NIH, Bethesda, MD
Area Chair, Medical Image Computing and Computer-Assisted Intervention (MICCAI)
Associate Editor, IEEE Transactions on Medical Imaging
Associate Editor, Medical Physics
Associate Editor, Medical Image Analysis
Associate Editor, Pattern Recognition

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Subject: Re: PE study to complete

[Ân văn bản trích dẫn]